

Teaching Exam

Subject – Computer Science

**Topic – Computer Organization &
Architecture**

Lecture No.- 11



By- Rahul Sir

Topics to be Covered

Topic



- 1:- Software and its type ✓
- 2:- Language Processor ✓
- 3:- Types of Language Processor ✓
- 4:- Encoding Schemes ✓
- 5:- Types of Encoding Schemes ✓
- 6:- Microprocessor – Uses, workflow, Application



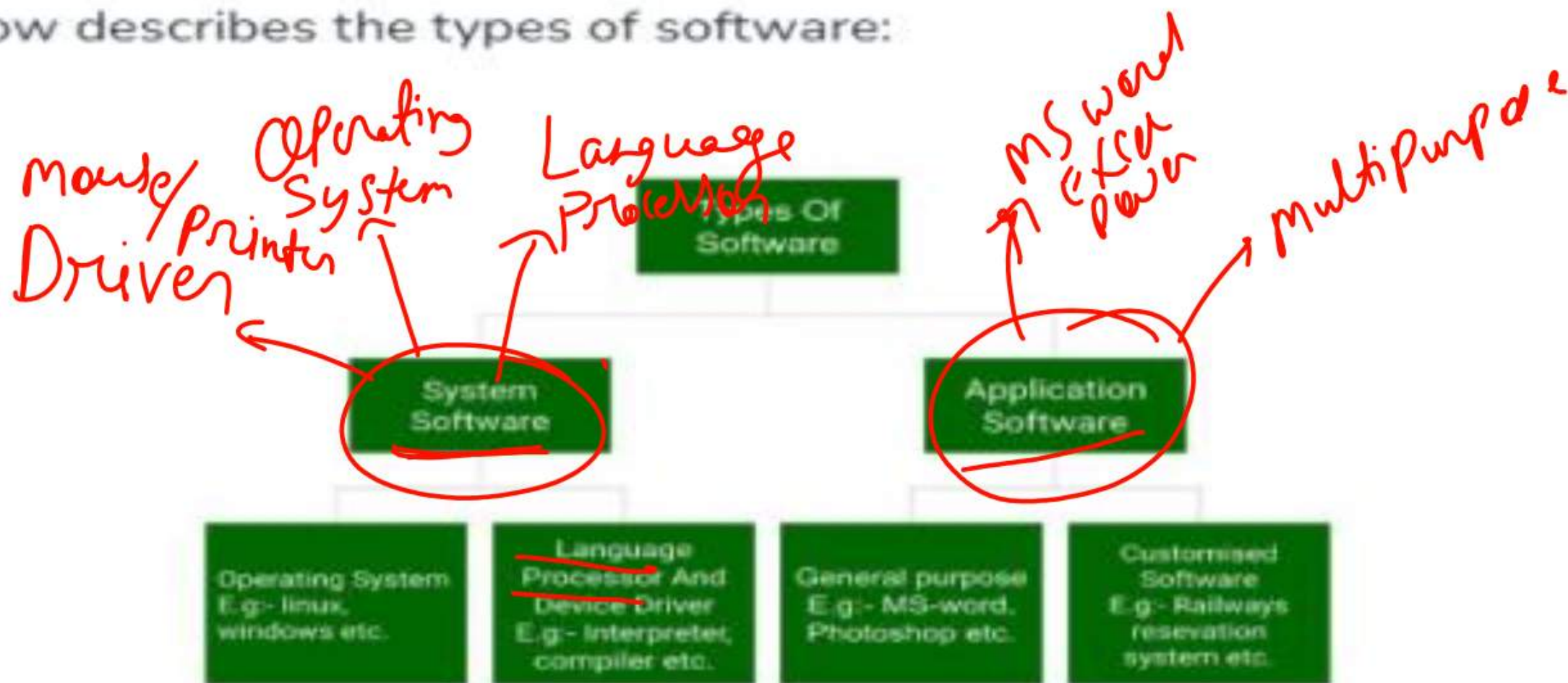
Topic: Software and its type

In a computer system, the software is basically a set of instructions or commands that tell a computer to do a specific task that serves its users.

- A software is not a physical thing like hardware, it rather makes the hardware work as per user requirements by giving instructions.
- Examples of software are MS-Word, MS-Excel, PowerPoint and Web Browser

Types of Software

The chart below describes the types of software:





Topic: System Software

1. System Software

- Operating System (like Windows and Linux)
- Language Processor
- Device Driver

2. Application Software

- General Purpose Software
- Customize Software
- Utility Software

System Software

System software is software that directly operates the computer hardware and provides the basic functionality to the users as well as to the other software to operate smoothly.

- System software basically controls a computer's internal functioning and also controls hardware devices such as monitors, printers, and storage devices, etc.
- It is like an interface between hardware and user applications, it helps them to communicate with each other because hardware understands machine language(i.e. 1 or 0) whereas user applications are work in human-readable languages like English, Hindi, German, etc.





Topic: Types of System Software

Types of System Software

It has two subtypes which are:

1. **Operating System:** It is the main program of a computer system. When the computer system ON it is the first software that loads into the computer's memory. Basically, it manages all the resources such as computer memory, CPU, printer, hard disk, etc., and provides an interface to the user, which helps the user to interact with the computer system.
2. **Language Processor:** As we know that system software converts the human-readable language into a machine language and vice versa. So, the conversion is done by the language processor. It converts programs written in high-level programming languages like Java, C, C++, Python, etc (known as source code), into sets of instructions that are easily readable by machines (known as object code or machine code).
(Handwritten red circles around Java, C, C++, and Python, with an arrow pointing to C++)
3. **Device Driver:** A device driver is a program or software that controls a device and helps that device to perform its functions. Every device like a printer, mouse, modem, etc. needs a driver to connect with the computer system eternally. So, when you connect a new device with your computer system, first you need to install the driver of that device so that your operating system knows how to control or manage that device.



Application Software

Software that performs special functions or provides functions that are much more than the basic operation of the computer is known as application software.

- Application software is designed to perform a specific task for end-users. It is a product or a program that is designed only to fulfill end-users' requirements.
- Examples include word processors, spreadsheets, database management, inventory, payroll programs, etc.



Types of Application Software

There are different types of application software and those are:

1. **General Purpose Software:** This type of application software is used for a variety of tasks and it is not limited to performing a specific task only. For example, MS-Word, MS-Excel, PowerPoint, etc.
2. **Customized Software:** This type of application software is used or designed to perform specific tasks or functions or designed for specific organizations. For example, railway reservation system, airline reservation system, invoice management system, etc.
3. **Utility Software:** This type of application software is used to support the computer infrastructure. It is designed to analyze, configure, optimize and maintains the system, and take care of its requirements as well. For example, antivirus, disk fragmenter, memory tester, disk repair, disk cleaners, registry cleaners, disk space analyzer, etc.



Topic: Difference between System and Application Software



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System Software	Application Software
It is designed to manage the resources of the computer system, like <u>memory</u> and <u>process management</u> , etc.	It is designed to fulfill the requirements of the user for performing <u>specific tasks</u> .
<u>Written in a low-level language.</u>	<u>Written in a high-level language.</u>
<u>Less interactive for the users.</u>	<u>More interactive for the users.</u>
<u>System software plays vital role for the effective functioning of a system.</u>	Application software is not so important for the functioning of the <u>system</u> , as it is task specific.
<u>It is independent of the application software to run.</u>	<u>It needs system software to run.</u>

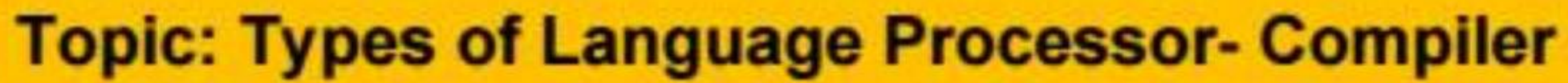


What is Language Processors?

Compilers, interpreters, translate programs written in high-level languages into machine code that a computer understands and assemblers translate programs written in low-level or assembly language into machine code. In the compilation process, there are several stages. To help programmers write error-free code, tools are available.

Assembly language is machine-dependent, yet mnemonics used to represent instructions in it are not directly understandable by machine and high-Level language is machine-independent. A computer understands instructions in machine code, i.e. in the form of 0s and 1s. It is a tedious task to write a computer program directly in machine code. The programs are written mostly in high-level languages like Java, C++, Python etc. and are called source code. These source code cannot be executed directly by the computer and must be converted into machine language to be executed. Hence, a special translator system software is used to translate the program written in a high-level language into machine code is called Language Processor and the program after translated into machine code (object program/object code).

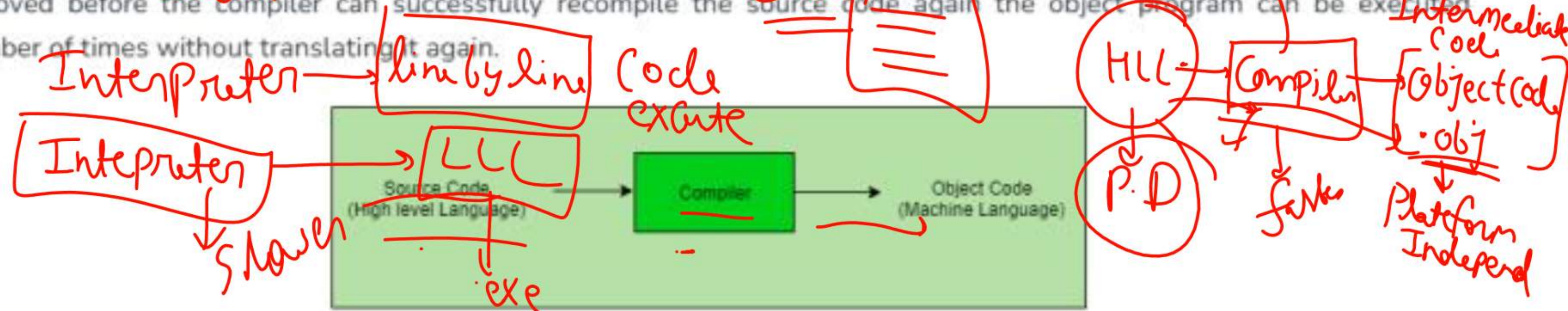
HLL → L.P. → LLL



The language processors can be any of the following three types:

The language processor that reads the complete source program written in high-level language as a whole in one go and translates it into an equivalent program in machine language is called a Compiler. Example: C, C++, C#.

In a compiler, the source code is translated to object code successfully if it is free of errors. The compiler specifies the errors at the end of the compilation with line numbers when there are any errors in the source code. The errors must be removed before the compiler can successfully recompile the source code again the object program can be executed number of times without translating it again.





Topic: Types of Language Processor- Assembler

2. Assembler

Language

ASSEMBLER

LLL

Addition
Subtraction
multiplication
DIV or

→ Mnemonic code
↓ (Capital letter
3 alphabets)

The Assembler is used to translate the program written in Assembly language into machine code. The source program is an input of an assembler that contains assembly language instructions. The output generated by the assembler is the object code or machine code understandable by the computer. Assembler is basically the 1st interface that is able to communicate humans with the machine. We need an assembler to fill the gap between humans and machine so that they can communicate with each other. code written in assembly language is some sort of mnemonics (instructions) like ADD, MUL, MUX, SUB, DIV, MOV and so on. and the assembler is basically able to convert these mnemonics in binary code. Here, these mnemonics also depend upon the architecture of the machine.

Addition

Subtraction

Multiplication

Division

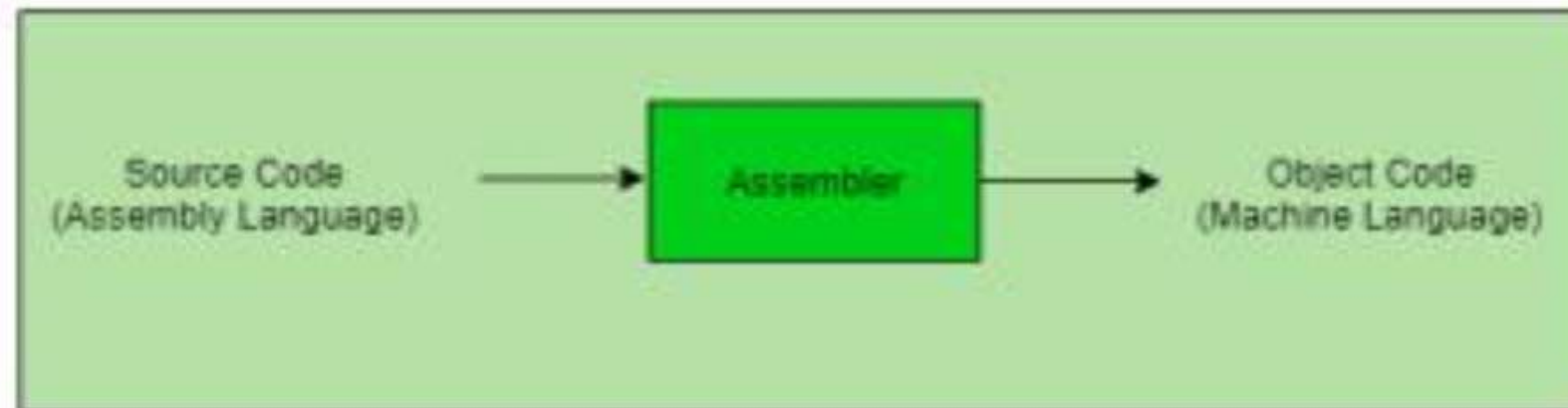
ADD

SUB

MUL

DIV

For example, the architecture of intel 8085 and intel 8086 are different.

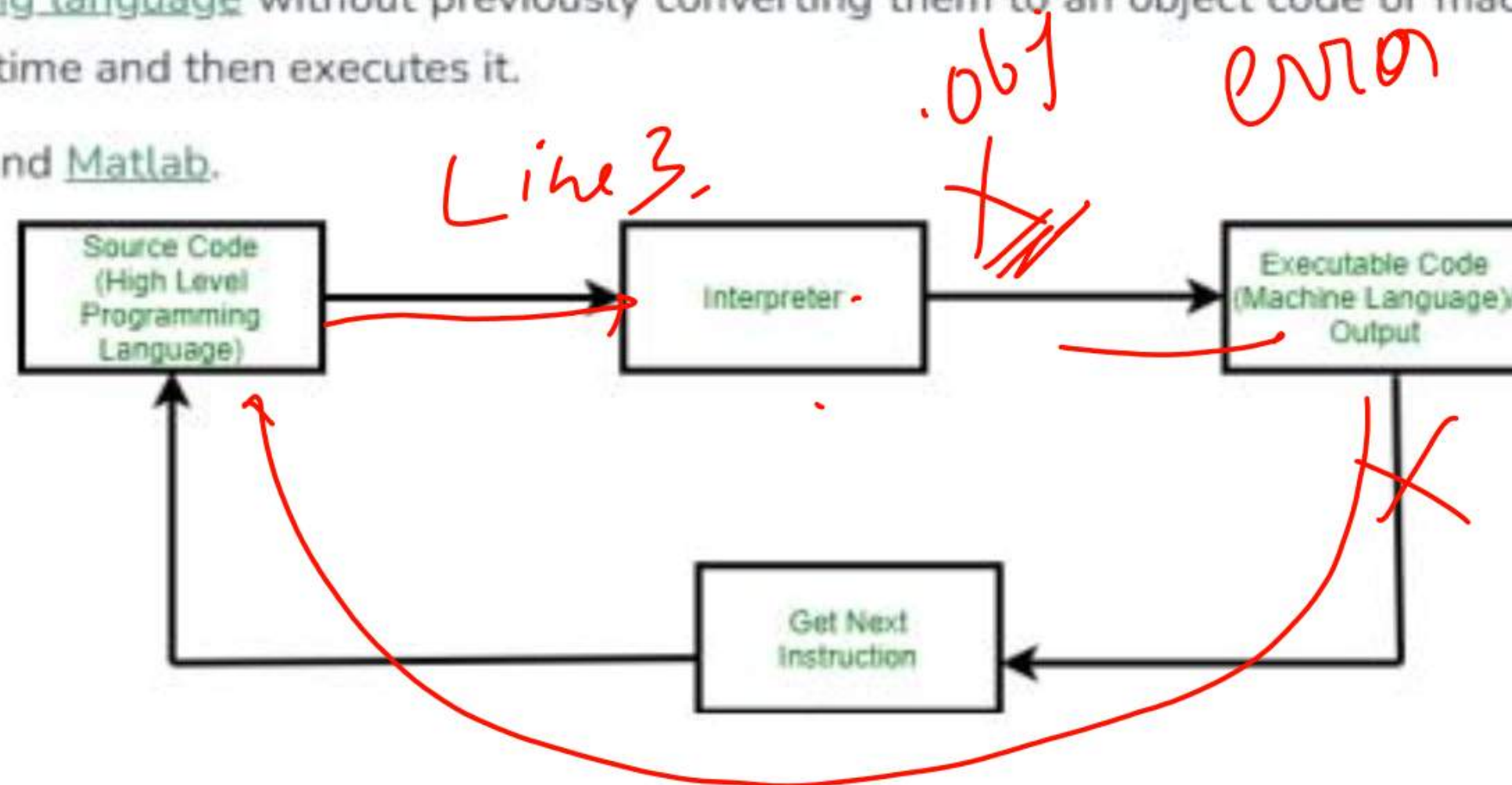




3. Interpreter

The translation of a single statement of the source program into machine code is done by a language processor and executes immediately before moving on to the next line is called an interpreter. If there is an error in the statement, the interpreter terminates its translating process at that statement and displays an error message. The interpreter moves on to the next line for execution only after the removal of the error. An Interpreter directly executes instructions written in a programming or scripting language without previously converting them to an object code or machine code. An interpreter translates one line at a time and then executes it.

Example: Perl, Python and Matlab.





Topic: Difference between Interpreter and Compiler



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.exe

Compiler	Interpreter
A compiler is a program that converts the entire source code of a <u>programming language</u> into executable machine code for a CPU.	An interpreter takes a source program and runs it line by line, translating each line as it comes to it.
The compiler takes a large amount of time to analyze the entire source code but the overall execution time of the program is comparatively faster.	An interpreter takes less amount of time to analyze the source code but the overall execution time of the program is slower.
The compiler generates the error message only after scanning the whole program, so debugging is comparatively hard as the error can be present anywhere in the program.	Its <u>Debugging</u> is easier as it continues translating the program until the error is met.
The compiler requires a lot of memory for generating object codes.	It requires less memory than a compiler because no object code is generated.
Generates intermediate object code.	No intermediate object code is generated.
For Security purpose compiler is more useful.	The interpreter is a little vulnerable in case of security.
Examples: C, C++, C#	Examples: Python, Perl, JavaScript, Ruby.

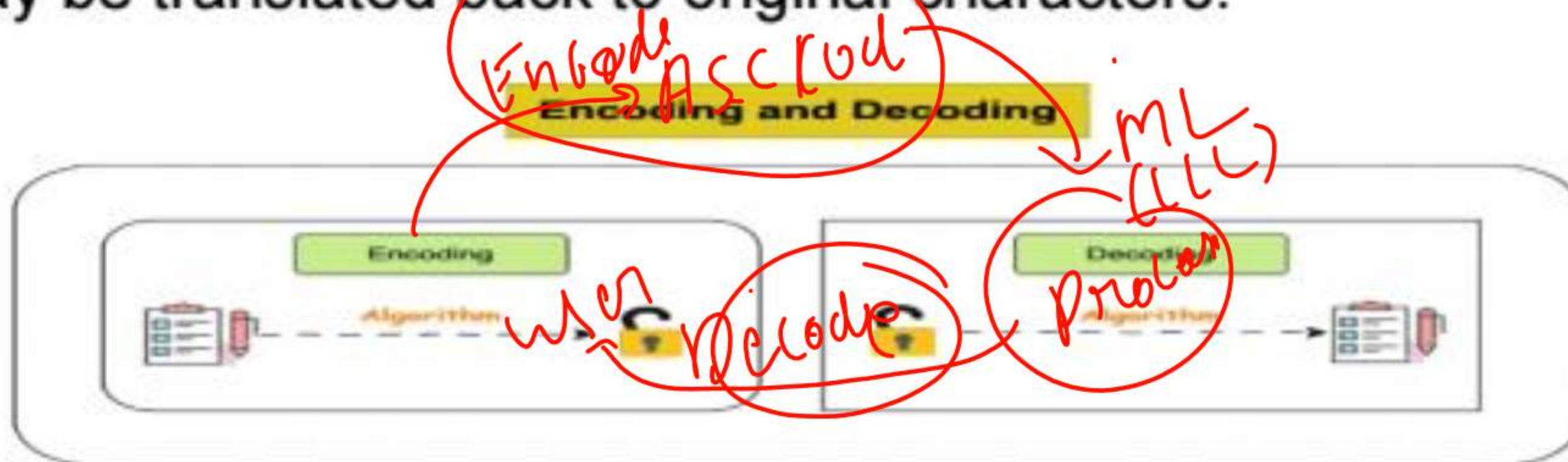
.exe



Encoding is a method of converting data from one format to another. When we have text (a series of characters) that we wish to keep inside a computer (machine) or transport over a digital network, we must convert it to binary representation since that is the only language that a binary-based computer can comprehend.

Character encoding is a method of converting text input to binary values. In a nutshell, we may give individual characters unique numeric values and transform those numbers to binary. Based on their values, these binary integers may be translated back to original characters.

$A = 65 \rightarrow \text{Binary}$

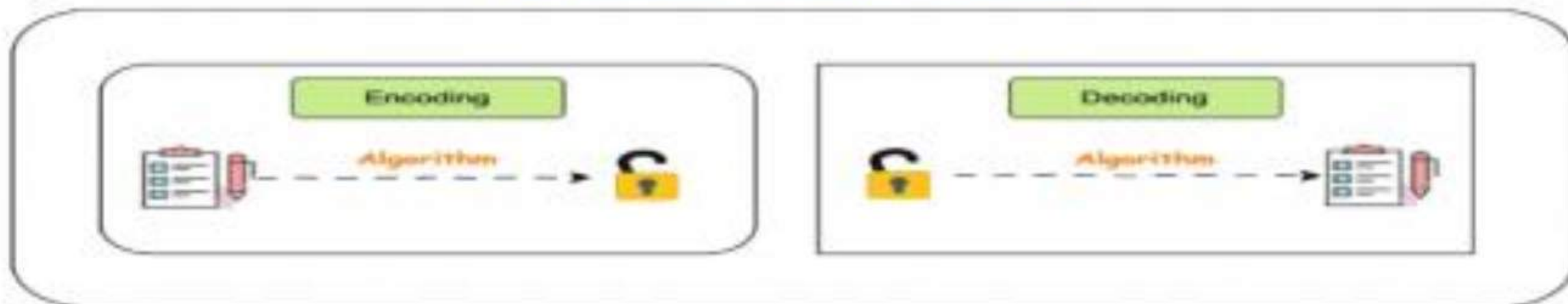




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Encoding and Decoding





Encoding schemes: Information Interchange (ASCII)

The American Standard Code for Information Interchange (ASCII) is a character encoding scheme that represents text in computers and other electronic devices. It uses a set of 128 characters, including letters, numbers, punctuation marks, and control characters, to represent text.

ASCII was first developed in the 1960s and is still widely used today, particularly in the United States and other parts of the world that use the Latin alphabet. It is the standard encoding scheme for text files in many operating systems and applications.

The ASCII character set consists of 128 characters, including:

- The uppercase and lowercase letters of the English alphabet (A-Z and a-z)
- The digits 0-9
- Punctuation marks and special characters, such as !, @, #, \$, %, &, etc.
- Control characters, such as the newline, tab, and backspace characters, which are used to control the formatting and layout of text.

ASCII is a 7-bit encoding scheme, which means that it uses 7 bits (or 1 byte) to represent each character. This allows for a total of 128 different characters to be represented. ASCII has been widely used for decades and still in use in some systems and protocols. However, it only supports the basic English alphabet and a limited set of special characters, which makes it not suitable for representing more complex characters, such as those used in other languages and scripts. Unicode and UTF-8 are more widely used encoding schemes that can represent a much larger set of characters.

ASCCod
↳ 128
↳ char
digit
Sym

$2^8 = 128$

a = 93

A = 65

ASCCod
7bit

A = 65

7bit (Binary)

Suppos
@ = 103
7bit



8 bit

256

Encoding schemes: Indian Script Code for Information Interchange (ISCII)

Indian Script Code for Information Interchange (ISCII) is a character encoding standard that is specifically designed to represent the scripts used in

India, such as Devanagari, Bengali, Gujarati, Gurmukhi, Kannada, Malayalam, Oriya, Tamil, Telugu, and Urdu. It was developed in the early 1990s by the Indian government's National Informatics Centre (NIC) and is still used in some systems and applications today.

ISCII is based on the 8-bit ASCII character set, which means that it uses 8 bits (or 1 byte) to represent each character. It includes a set of characters for each of the Indian scripts, as well as additional characters for punctuation marks, diacritics, and other special characters.



Topic: Encoding Schemes - ISCII

ISCII is based on the 8-bit ASCII character set, which means that it uses 8 bits (or 1 byte) to represent each character. It includes a set of characters for each of the Indian scripts, as well as additional characters for punctuation marks, diacritics, and other special characters.

The ISCII character set consists of several different code pages, each of which is designed for a specific script. The code pages include:

- Devanagari script code page
- Bengali script code page
- Gujarati script code page
- Gurmukhi script code page
- Kannada script code page
- Malayalam script code page
- Oriya script code page
- Tamil script code page
- Telugu script code page
- Urdu script code page

ISCII has been widely used in India for decades, but it has some limitations, as it can't handle complex scripts and it's not compatible with other encoding standards, like Unicode. Unicode is now more widely used to represent Indian scripts, as it can handle a much larger set of characters and it's more widely supported. However, some legacy systems still use ISCII encoding and many Indian Language Processing Applications still use ISCII for backward compatibility.



Encoding schemes: Unicode

Unicode is a character encoding standard that represents text in computers and other electronic devices. It is a universal character set that includes almost every character from every writing system in the world, including letters, numbers, punctuation marks, and symbols.

Unicode was first developed in the late 1980s and has since become the standard character encoding scheme for text files in many operating systems and applications. It is designed to be backward-compatible with ASCII and extends the number of characters that can be represented from 128 to over 130,000.

The Unicode character set consists of a vast number of characters, including:

- Characters from the basic Latin alphabet (A-Z and a-z) UT
- Characters from other scripts such as Greek, Chinese, Japanese, Arabic, etc.
- Mathematical symbols, emoji and various types of punctuation marks
- Control characters, such as the newline, tab, and backspace characters, which are used to control the formatting and layout of text.

UTF → 8, UTF-16

UTF

32

32 bit

2³²



Unicode uses a variable-width encoding which means that it uses a different number of bytes to represent different characters. Characters that are used more frequently, such as the basic Latin alphabet, are encoded using fewer bytes, while less common characters, such as those from non-Latin scripts, are encoded using more bytes. Unicode can be encoded using different formats, the most common formats are UTF-8, UTF-16, and UTF-32.

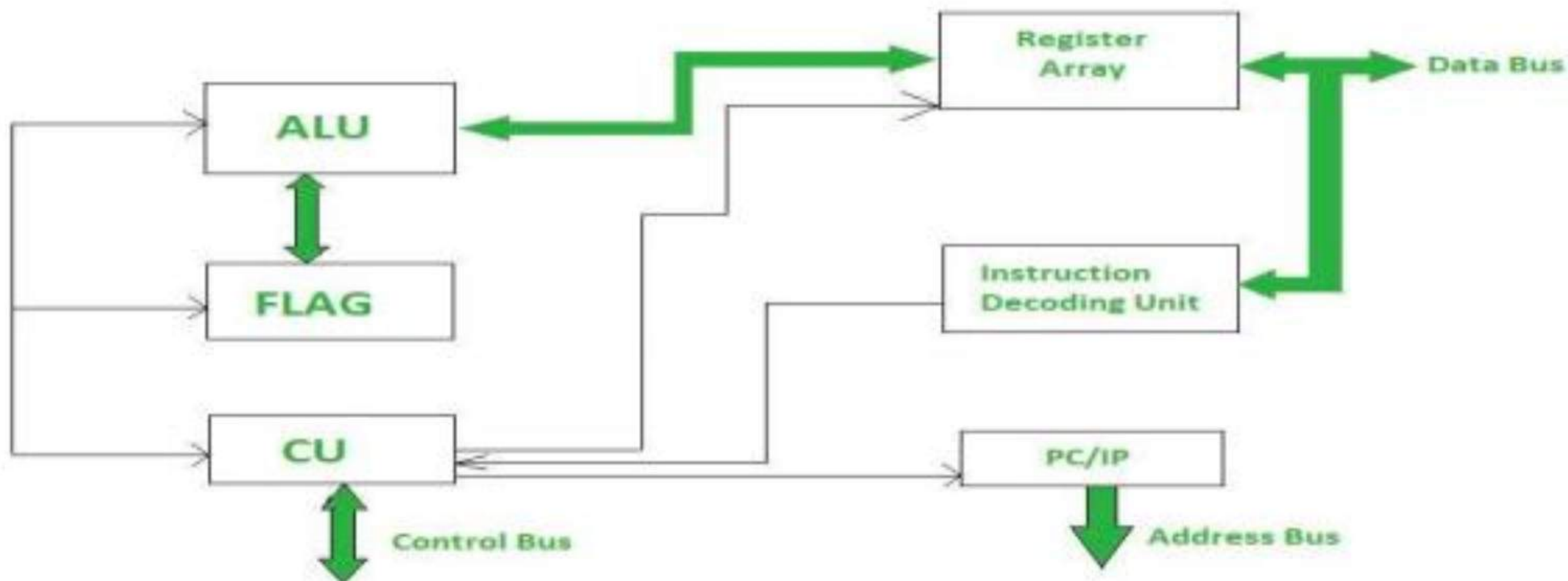
UTF-8 is an encoding of Unicode that uses 8-bit sequences to represent characters, it is backward-compatible with ASCII and it's the most popular encoding format on the internet. UTF-16 uses 16-bit sequences to represent characters, it's mostly used in systems that need to handle a large number of characters, such as Windows. UTF-32 uses 32-bit sequences to represent characters, it's used in systems that need to handle a large number of characters and it's more efficient than UTF-16 in terms of memory usage.



What is Microprocessor?

A Microprocessor is a programmable device that takes in input, performs some arithmetic and logical operations over it and produces the desired output. In simple words, a Microprocessor is a digital device on a chip that can fetch instructions from memory, decode and execute them, and give results. It is an important part of a computer architecture without which you will not be able to perform anything on your computer.

A typical Microprocessor structure looks like this.





Block Diagram of a Microprocessor

A Microprocessor takes a bunch of instructions in machine language and executes them, telling the processor what it has to do. The microprocessor performs three basic things while executing the instruction:

- It performs some basic operations like addition, subtraction, multiplication, division, and some logical operations using its Arithmetic and Logical Unit ([ALU](#)). New Microprocessors also perform operations on floating-point numbers.
- Data in microprocessors can move from one location to another.
- It has a [Program Counter](#) (PC) register that stores the address of the next instruction based on the value of the PC, Microprocessor jumps from one location to another and makes decisions.



Basic Terms used in Microprocessor

Given below are the some basic term used in the microprocessors

- **Instruction Set:** Instruction set is a group of instructions in the microprocess which can be executed. It is the Interface between hardware and software.
- **Bus:** Buses are used to carry data, address and control information within a microprocessor. There are three types of Bus which are data buses, address buses, and control buses.
- **IPC (Instructions Per Cycle):** It is Measured as the number of instructions that a CPU can execute in a Single clock cycle.
- **Clock Speed:** It refers to the number of operations a processor can perform per second. It is Measured in megahertz (MHz) or gigahertz (GHz).
- **Bandwidth:** It is measured as is the number of bits processed during a single instruction.
- **Word Length :**It Shows the number of bits a processor can handle at a time.
- **Data Types:** It Supports various formats such as binary, ASCII, Signed and Unsigned Bits.



Evolution of Microprocessors

The Evolution of Microprocessors are categorize as

- **First Generation (4-bit Microprocessors):** The Intel Corporation first introduced the microprocessor in 1971-1972, naming it the Intel 4004. This initial microprocessor could perform basic arithmetic and logical functions. It also contained a control unit capable of fetching, decoding, and executing instructions.
- **Second Generation (8-bit Microprocessor):** The intel corporation first introduced the 8 bit microprocessor in 1973 with the intel 8008. This was followed by an enhanced model, the intel 8088.
- **Third Generation (16-bit Microprocessors):** The Third Generation introduced microprocessors capable of performing 16 bit operations. Introduced in 1978, this generation was represented by processors like the Intel 8086, Zilog Z800 and 80286.
- **Fourth Generation (32 - bit Microprocessors):** The most Used and famous 32 bit microprocessor was intel 80386.
- **Fifth Generation (64 - bit Microprocessors):** Since 1995, microprocessors with 64-bit processors which features 64 bit processors like Intel's Pentium pro which allows multiple CPU's in one.



Types of Processor

Complex Instruction Set Computer (CISC)

CISC or Complex Instruction Set Computer is a computer architecture where instructions are such that a single instruction can execute multiple low-level operations like loading from memory, storing into memory, or an [arithmetic operation](#), etc. It has multiple addressing modes within a single instruction. CISC makes use of very few registers.

Example of CISC are

- Intel 386
- Intel 486
- Pentium
- Pentium Pro
- Pentium II
- Pentium III
- Motorola 68000
- Motorola 68020
- Motorola 68040 etc.



Reduced Instruction Set Computer (RISC)

RISC or Reduced Instruction Set Computer is a computer architecture where instruction is simple and designed to get executed quickly. Instructions get completed in one clock cycle this is because of the optimization of instructions and pipelining (a technique that allows for simultaneous execution of parts, or stages, of instructions more efficiently process instructions). RISC makes use of multiple registers to avoid large interactions with memory. It has few addressing nodes.

Example of RISC are

- IBM RS6000
- MC88100
- DEC Alpha 21064
- DEC Alpha 21164
- DEC Alpha 21264

Explicitly Parallel Instruction Computing (EPIC)

EPIC or Explicitly Parallel Instruction Computing permits computers to execute instructions parallel using compilers. It allows complex instructions execution without using higher clock frequencies. EPIC encodes its instruction into 128-bit bundles. each bundle contains three instructions which are encoded in 41 bits each and a 5-bit template field (contains information about types of instructions in a bundle and which instructions can be executed in parallel).

Example of EPIC is

- IA-64 (Intel Architecture-64)



Applications of Microprocessor

Given below are the Applications of the Microprocessors

- **Computer:** The Microprocessors are the CPU's in computers.
- **Embedded Systems:** It is utilized as the main processing block in the Embedded systems such as Washing machines, microwaves and other equipments.
- **Industrial Automation:** It can be used to control Industrial machinery and equipments.
- **Automotive:** The Modern day Vehicles uses Microprocessors in there ECUs.
- **Telecommunications:** The Microprocessors are utilized in the Telecommunications systems such as routers, modems, and switches.

THANK YOU